Predicting Bike Rental Demand Based on Weather and Time

Course: AI 101 Mini-Project | Team 15
Repo: github.com/2026eeb1330/Bike-Rental-Demand-Prediction

Team Members & Roles:
• Vantala Venu Prathyush — Execution Lead
• Avishi Sood — Documentation Lead
• Mirgane Aditya Tanaji — Video Lead
• Pooja (2026MCB1554) — PPT Lead

Problem Statement & Dataset

- Problem Statement: Weather aur time-based factors ka use karke daily bike rental demand predict karna.
- Dataset: Bike Sharing Dataset (data/bike_sharing.csv)
- Key Features: temp, humidity, windspeed, season, workingday, weather
- Target Variable: count (Total Bike Rentals)

Pipeline & Methodology

- Workflow: Data Preprocessing -> EDA -> Model Training -> Evaluation
- Models Used: Linear Regression & Decision Tree Regressor
- Libraries Used: Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn

Results & Model Performance

- Linear Regression: R2 Score = 0.78 | RMSE = 102.3 | MAE = 75.4
- Decision Tree: R2 Score = 0.85 | RMSE = 84.1 | MAE = 58.2

Learnings & AI Disclosure

- Key Learnings: Model evaluation metrics analysis & Git workflow.
- Team Roles: Execution, Documentation, PPT Lead, Video Lead.
- AI Disclosure: ChatGPT / Gemini were used for syntax debugging and structuring documentation.